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The relationship between trading volume, volatility and returns of Non-Fungible Tokens: evidence from a quantile approach

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ABSTRACT

This is the first study to examine the quantile connectedness for returns-volume and volatility-volume pairs for the three non-fungible tokens (THETA, Tezos, and Enjin Coin) using the quantile VAR approach. The results report the highest connectedness of volume with returns and volatility in the extreme upper quantile compared to other quantiles, implying the asymmetric connectedness. The spillover effect is observed from volume to returns and volatilities in extreme upper and lower market conditions, whereas opposite direction of spillovers is evident for the selected non-fungible tokens at median quantile. Our findings are useful for investors in predicting the returns and risk of NFTs using trading volume in the extreme market conditions.

C12, C22, G14, G15

1. Introduction

How can a piece of digital art be worth several millions of dollars? It has become possible since the introduction of so-called Non-Fungible tokens (NFTs). NFTs are unique digital tokens representing the ownership of certain unique digital objects, e.g., artworks, images, videos or similar. The astonishing popularity of conventional cryptocurrencies, such as Bitcoin and Ethereum, has attracted the attention of investors to new and non-interchangeable type of digital assets. While a wide variety of digital assets have been introduced in the last decade, disrupting global financial markets in terms of payments, investments and financial services (E.g., Böhme et al., 2015; Cong et al., 2021; Ante et al., 2021), NFTs are the most recent innovation and relatively under-researched investment instrument to date (e.g., Dowling, 2021a; Kireyev and Evans, 2021; Yousaf and Yarovaya, 2022; Baals et al., 2022).

The prices and trading volume of NFTs have dramatically increased in the last three years, i.e., the total trading volume of THETA was $3.31 billion in first semi-annual of 2018 and it increased to $74.12 billion in first semi-annual of 2021. Moreover, the price of THETA was increased from $0.18 on 17 January 2018 to $6.48 on 20 November 2021. There is a variation in price and volume growth over time as well in the THETA. These prices and volume hikes were also observed in other NFTs blockchains (i.e., XTZ and ENJ) but with higher volatility as well (Wilson et al., 2021). Specifically, Ante (2021) reports that the trading volume of NFTs increase 23 folds from 2018 to 2021. These unprecedented hikes of prices, volume, and their associated volatility motivate us to examine the linkages among them. A very important research question is that whether volume affect the returns and volatility or vice versa? These directions of relationships are important in understanding the information arrival process, price spikes, and crashes (Marsh and Wagner, 2004).

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Fig. 1. Prices, returns, volatility, volume, and De-trended Volume.

Table 2
Static quantile-based spillovers between return and volume.

	Median (0.50)	Lower Quantile (0.05)	Upper Quantile (0.95)
1. THETA			
Total spillover	14.01	38.86	40.36
To:			
Returns	14.5	38.4	40.64
Volume	13.49	39.33	40.08
Net:			
Returns	1.01	-0.93	0.55
Volume	-1.01	0.93	-0.55
2. XTZ			
Total spillover	5.82	36.42	38.27
To:			
Returns	6.70	35.72	37.97
Volume	4.95	37.12	38.58
Net:			
Returns	1.75	-1.40	-0.62
Volume	-1.75	1.40	0.62
3. ENJ			
Total spillover	14.26	37.44	40.92
To:			
Returns	14.77	37.19	40.14
Volume	13.76	37.68	41.7
Net:			
Returns	1.01	-0.50	-1.55
Volume	-1.01	0.50	1.55

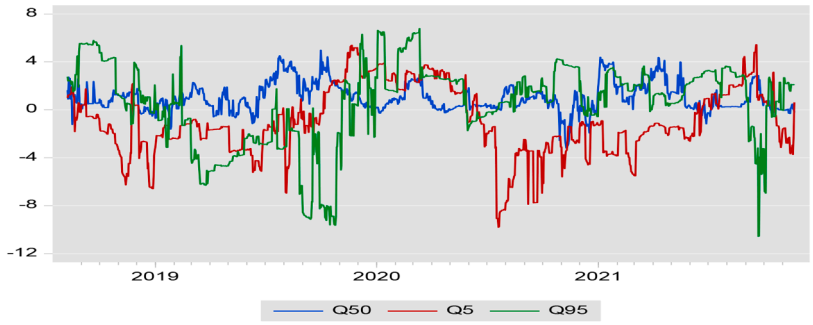
Notes: 'Total spillover' means the total connectedness between return and volume of NFT. 'To returns' indicates the spillover from volume to returns, whereas 'To Volume' represents the spillover from return to volume. The positive 'Net returns' spillover shows the returns as net transmitter of spillover to volume, and vice versa. The length of rolling window is 200 days.

Table 3
Static quantile-based spillovers between volatility and volume.

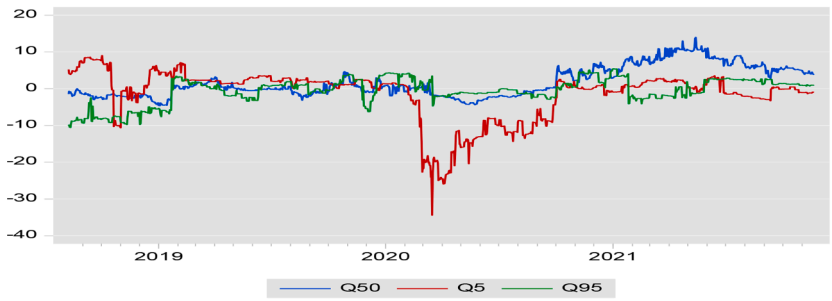
	Median (0.50)	Lower Quantile (0.05)	Upper Quantile (0.95)
1. THETA			
Total spillover	13.03	18.41	33.8
To:			
Volatility	12.56	17.89	30.72
Volume	13.49	18.93	36.88
Net:			
Volatility	-0.93	-1.05	-6.16
Volume	0.93	1.05	6.16
2. XTZ			
Total spillover	12.38	20.51	41.15
To:			
Volatility	11.91	20.2	40.84
Volume	12.84	20.83	41.46
Net:			
Volatility	-0.93	-0.63	-0.62
Volume	0.93	0.63	0.62
3. ENJ			
Total spillover	12.68	17.2	30.17
To:			
Volatility	12.76	17.09	26.58
Volume	12.60	17.31	33.76
Net:			
Volatility	0.17	-0.22	-7.18
Volume	-0.17	0.22	7.18

Notes: 'Total spillover' means the total connectedness between volatility and volume of NFT. 'To volatility' indicates the spillover from volume to volatility, whereas 'To Volume' represents the spillover from volatility to volume. The positive 'Net volatility' spillover shows the volatility as net transmitter of spillover to volume, and vice versa. The length of rolling window is 200 days.

a) THETA (Returns-Volume)



b) XTZ (Returns-Volume)



c) ENJ (Returns-Volume)

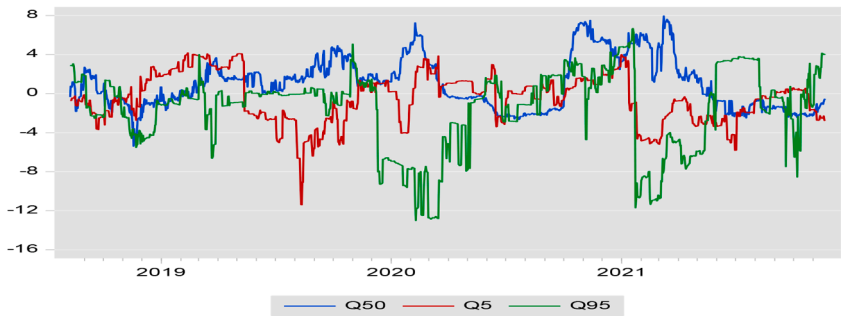


Fig. 2. Time-varying net spillovers in quantile VAR for returns-volume pairs of NFTs [The positive value indicates that returns are the net transmitter of spillover to volume whereas negative value shows that returns are net recipient of spillover from volume].

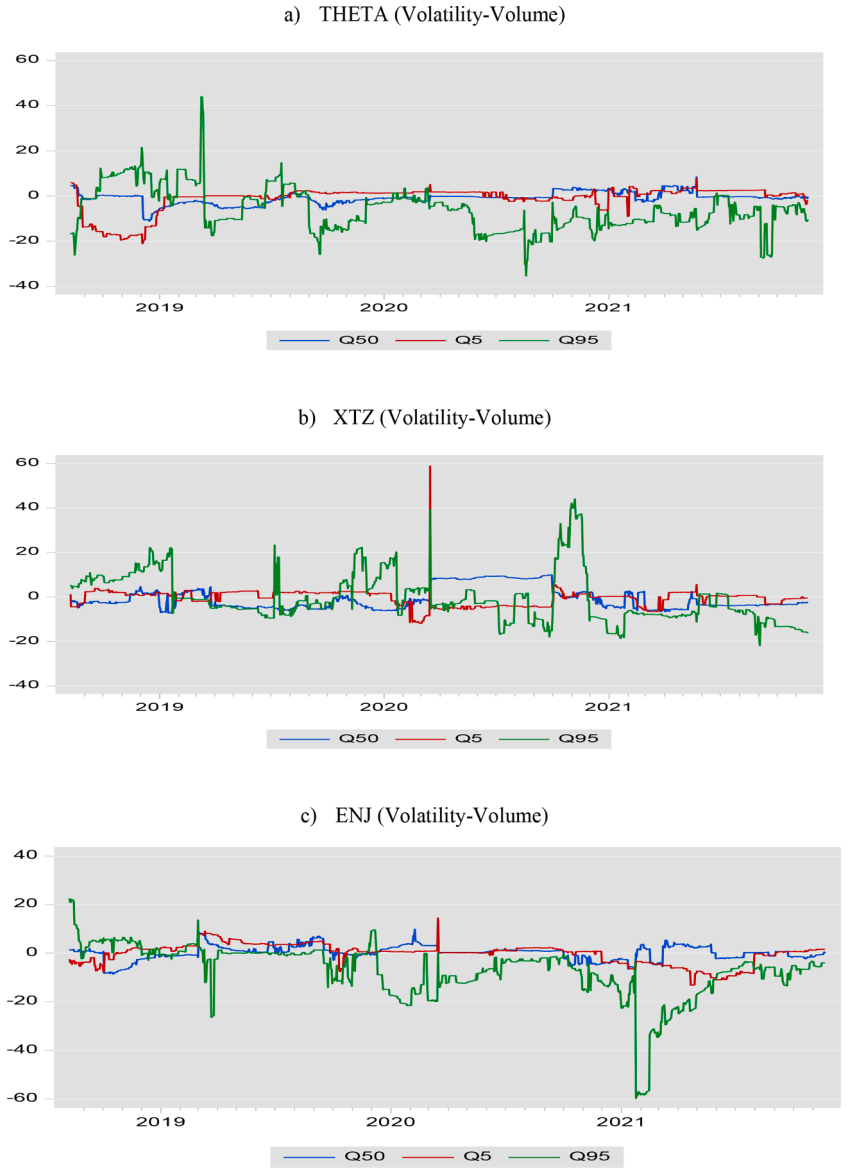


Fig. 3. Time-varying net spillovers in quantile VAR for volatility-volume pairs of NFTs [The positive value indicates that volatility is the net transmitter of spillover to volume whereas negative value shows that volatility is net recipient of spillover from volume].

between "TO" and "FROM," the negative (positive) value refers to the net recipient (transmitter) of spillover. TCI_t represents average level of total connectedness.

3. Empirical results

Our results provide three important pieces of evidence. First, according to the results from static quantile-based spillover analysis presented in Table 2, the total connectedness between return and volume is the highest in the upper quantile, whereas lowest in the median quantile for all three NFTs. These results indicate the asymmetric spillovers between return and volume, suggesting that investors should formulate different strategies during normal and extreme bullish and extreme bearish market conditions of NFTs. These findings are in line with Naeem et al. (2020) who also provide the evidence of asymmetric tail connectedness between volume and returns of fungible cryptocurrencies, like Bitcoin, Ethereum, and Litecoin. The net spillover of returns is positive at median quantile in three NFTs, showing that the direction of spillovers is from return to volume in three NFTs analyzed. However, the net spillover of returns is negative at extreme lower (upper) quantile in three NFTs (XTZ and ENJ), therefore the direction of spillover is from volume to returns in extreme upper and lower quantiles, in line with anti-efficiency market hypothesis. These results suggest that the volume has predictive power for returns only in extreme bullish and bearish market conditions, which indicates the NFT market as inefficient

market. This predictive power of volume for returns in extreme market conditions is also useful to make trading decisions. Bouri et al. (2019) report the volume as predictor of positive and negative returns in leading (fungible) cryptocurrencies. Further, volume is not the predictor for returns in the extreme bullish market condition for THETA because the net spillover of returns is positive for THETA in upper quantile.

Second, the results of static quantile-based spillovers between volatility and volume presented in Table 3 show that the total connectedness index is higher in upper quantile compared to the lower and median quantiles, providing the evidence of asymmetric connectedness between volatility and volume of three NFTs. At median quantile, the net spillover of volatility is negative in THETA and XTZ but positive in ENJ, indicating that volume can be used to forecast volatility of THETA and XTZ in normal or average market conditions. Bouri et al. (2019) report that volume can forecast volatility of only few traditional fungible cryptocurrencies, and our results add to their findings providing novel evidence from non-fungible cryptocurrencies. In the upper and lower quantiles, the net spillover of volatility is negative in three NFTs therefore volatility is the net recipient of spillover from volume, implying that the information content of volume is useful in forecasting the volatility of three NFTs in extreme bearish and extreme bullish market conditions. These findings are in line with anti-efficiency market hypothesis and sequential information arrival hypothesis. We check the robustness of static results by changing the sample period[3] and empirical method[4], see Table A1, A2, and A3.

Third, we illustrate the time-varying spillover between returns and volume for three NFTs in Fig. 2, and show that returns are net transmitters to volume throughout the whole observation period in THETA and ENJ, whereas returns were net recipient (transmitters) of spillover from (to) volume before (after) third quarter of 2020 only. In extreme upper and lower quantiles, the status of returns as net recipient or net transmitter varies frequently over time, which is similar to the findings of Naeem et al. (2019) in traditional cryptocurrencies. Fig. 3 demonstrates that volatility is net recipient of spillover from volume before mid of 2020 in THETA whereas patterns of net spillovers are mixed after that time at median quantile. The status of returns as net recipient or transmitter frequently varies in XTZ and ENJ. In extreme bullish conditions, volatility is net recipients of spillover from volume over majority sample period in THETA and ENJ, whereas the patterns of extreme upper spillovers are mixed in XTZ over time. The net spillovers of returns frequently change in extreme lower quantile in three NFTs. To summarize, the frequent change in status of returns/volatility as net recipient or transmitter of spillover suggests that investors of NFTs should continuously monitor the movements of market's trading volume and prices and adjust the investment strategy over time to get higher returns.

4. Conclusion

Previous studies have only focused on the examining of the return-volume or volatility-volume nexus of fungible tokens, such as Bitcoin and other 'conventional' cryptocurrencies. Our results add to a better understanding of return-volume and volatility-volume relationships of the non-fungible tokens, i.e. THETA, Tezos, and Enjin Coin that are built on alternative blockchain to increasingly popular among NFT communities Ethereum protocol. Using Quantile VAR approach and daily data for three popular NFTs for the period from 17 January 2018 to 20 November 2021, we reveal that trading volume is strongly connected to the returns and volatilities at extreme bullish market conditions compared to the other quantiles. The results report the asymmetric connectedness of trading volume with returns and volatilities, where connectedness is also time varying. These findings are useful for cryptocurrency scholars aiming to understand the difference in return-volume and volatility-volume relationships among different types of digital assets. In terms of future research directions, we suggest comparing the return-volume and/or volatility-volume nexus for traditional cryptocurrencies, NFTs, and Defi assets simultaneously.

Author statement

Imran Yousaf: data collection; empirical analysis; performing robustness tests; writing.
Larisa Yarovaya: conceptualisation; literature review; writing and editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Tables A1, A2 and A3

[3] We use the sample period from 17 February 2018 and ends on 20 November 2021, and calculate the spillovers at median, extreme upper and extreme lower quantiles.

[4] We use the Diebold and Yilmaz (2012) approach to calculate the mean-based spillovers, which are only comparable to the median (Q=0.50) quantile.

Table A1
Static quantile-based spillovers between return and volume using sample period from 17 February 2018 to 20 November 2021.

	Median (0.50)	Lower Quantile (0.05)	Upper Quantile (0.95)
1. THETA			
Total spillover	13.83	38.79	40.27
To:			
Returns	14.34	38.31	40.53
Volume	13.33	39.27	40.01
Net:			
Returns	1.01	-0.96	0.52
Volume	-1.01	0.96	-0.52
2. XTZ			
Total spillover	5.88	36.35	38.39
To:			
Returns	6.78	35.59	38.15
Volume	4.98	37.11	38.63
Net:			
Returns	1.80	-1.51	-0.47
Volume	-1.80	1.51	0.47
3. ENJ			
Total spillover	14.39	37.45	41.00
To:			
Returns	14.90	37.21	40.19
Volume	13.88	37.70	41.80
Net:			
Returns	1.03	-0.49	-1.61
Volume	-1.03	0.49	1.61

Notes: 'Total spillover' means the total connectedness between return and volume of NFT. 'To returns' indicates the spillover from volume to returns, whereas 'To Volume' represents the spillover from return to volume. The positive 'Net returns' spillover shows the returns as net transmitter of spillover to volume, whereas negative 'Net returns' spillover indicate the returns as net recipient of spillover from volume. The length of rolling window is 200 days.

Table A2
Static quantile-based spillovers between volatility and volume using sample period from 17 February 2018 to 20 November 2021.

	Median (0.50)	Lower Quantile (0.05)	Upper Quantile (0.95)
1. THETA			
Total spillover	13.01	18.34	33.70
To:			
Volatility	12.52	17.80	30.68
Volume	13.51	18.88	36.71
Net:			
Volatility	-0.99	-1.09	-6.03
Volume	0.99	1.09	6.03
2. XTZ			
Total spillover	12.28	20.44	41.00
To:			
Volatility	11.83	20.15	40.64
Volume	12.73	20.73	41.36
Net:			
Volatility	-0.90	-0.58	-0.72
Volume	0.90	0.58	0.72
3. ENJ			
Total spillover	12.85	17.26	30.13
To:			
Volatility	12.93	17.18	26.37
Volume	12.77	17.33	33.90
Net:			
Volatility	0.16	-0.16	-7.53
Volume	-0.16	0.16	7.53

Notes: 'Total spillover' means the total connectedness between volatility and volume of NFT. 'To volatility' indicates the spillover from volume to volatility, whereas 'To Volume' represents the spillover from volatility to volume. The positive 'Net volatility' spillover shows the volatility as net transmitter of spillover to volume, whereas negative 'Net volatility' spillover indicate the volatility as net recipient of spillover from volume. The length of rolling window is 200 days.

Table A3
Static mean-based spillovers for the return-volume and volatility-volume using the Diebold and Yilmaz (2012) model.

	Return-volume		Volatility-volume
1. THETA		**1. THETA**	
Total spillover	14.41	**Total spillover**	12.96
To:		**To:**	
Return	15.12	Volatility	11.80
Volume	13.70	Volume	14.11
Net:		**Net:**	
Return	1.42	Volatility	-2.30
Volume	-1.42	Volume	2.30
2. XTZ		**2. XTZ**	
Total spillover	5.86	**Total spillover**	14.01
To:		**To:**	
Return	6.92	Volatility	13.87
Volume	4.81	Volume	14.14
Net:		**Net:**	
Return	2.11	Volatility	-0.26
Volume	-2.11	Volume	0.26
3. ENJ		**3. ENJ**	
Total spillover	14.72	**Total spillover**	12.06
To:		**To:**	
Return	15.63	Volatility	12.72
Volume	13.82	Volume	11.40
Net:		**Net:**	
Return	1.81	Volatility	1.31
Volume	-1.81	Volume	-1.31

Notes: 'Total spillover' indicatess the total connectedness between Return/volatility and volume of NFT. 'To returns' indicates the spillover from volume to returns, whereas 'To Volume' represents the spillover from return to volume. 'To volatility' indicates the spillover from volume to volatility, whereas 'To Volume' represents the spillover from volatility to volume. The positive 'Net returns' spillover shows the returns as net transmitter of spillover to volume, whereas negative 'Net returns' spillover indicate the returns as net recipient of spillover from volume. The positive 'Net volatility' spillover shows the volatility as net transmitter of spillover to volume, whereas negative 'Net volatility' spillover indicate the volatility as net recipient of spillover from volume. The length of rolling window is 200 days.

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